

B.TECH.
(SEM IV) EVEN SEMESTER EXAMINATION 2023-24
MATERIAL SCIENCE

[TIME: 3 hrs.]

[Max. Marks: 70]

Note: Attempt All Questions. All Question carry equal marks.

- | | | |
|------------|--|-------|
| Q1. | Answer ALL parts. | Marks |
| (a) | State the Hume-Rothery rules that favours extensive substitutional solid solubility. | 7 |
| (b) | Explain Gibb's phase rule. Determine the degree of freedom for an isomorphous alloy system when both the phases co-exist at equilibrium. | 7 |

OR

Define phase. Discuss the binary phase diagram for isomorphous system using lever rule.

- | | | |
|------------|---|---|
| Q2. | Answer ALL parts. | |
| (a) | Explain Iron-Carbon (Fe-C) equilibrium diagram with neat sketch. Briefly explain the various reactions and phases present in the diagram. | 7 |
| (b) | Explain TTT diagram for eutectoid steel and describe the effect of cooling rate on transformation product and hardness obtained. | 7 |

OR

Explain the properties and use of different types of stainless steels commonly used in industries.

- | | | |
|------------|--|---|
| Q3. | Answer ALL parts. | |
| (a) | Define failure and fracture. Explain the different modes of fracture in metals with neat sketch. | 7 |
| (b) | Discuss in brief the different mechanism of strengthening in metals and alloys. | 7 |

OR

What is hardness testing? How does micro hardness testing differ from macro hardness testing, and what are some common techniques used in hardness testing?

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|------------|--|---|
| Q4. | Answer ALL parts. | |
| (a) | What do you understand by ferromagnetism, anti-ferromagnetism and ferrimagnetism? Explain domain theory of ferromagnetism. | 7 |
| (b) | Explain retentivity and coercivity in hysteresis curve. Discuss some applications of hysteresis curve. | 7 |

OR

Discuss the types of polarization and also talk about the frequency effects on polarization.

- Q5.** Answer **ALL** parts. 7
- (a) How composites materials are superior than traditional engineering alloys? Classify composites materials based on reinforcement type. 7
- (b) Explain ceramic materials with suitable examples. What are the different properties and applications of ceramic materials?

OR

Describe nanomaterials. Describe a few uses for nanotechnology. How nanomaterials are categorized according to dimensions?

Subject code: IAS 401

Roll No: 2200520100061

B.TECH.
(SEM IV) EVEN SEMESTER EXAMINATION 2023-24
TECHNICAL COMMUNICATION

[TIME: 3 hrs.]

[Max. Marks: 70]

Note: Attempt All Questions. All Question carry equal marks.

- | | Marks |
|---|-------|
| Q1. Answer ALL parts. | |
| (a) Discuss the demerits of channel /flow of communication in professional world. | 3.5 |
| (b) Discuss the process of communication. | 3.5 |
| OR | |
| Define the term syllables, phonemes | |
| (c) Explain Intrapersonal, interpersonal, mass communication with examples. | 3.5 |
| OR | |
| Draw a syllabic structure of the following words: noun, adverb, fellow, thirsty, before. | |
| (d) Discuss the importance of Voice quality during your presentation. | 3.5 |
| Q2. Answer ALL parts. | |
| (a) Discuss barriers to communication in details with examples. | 07 |
| (b) Discuss the importance of group discussion and 05 do's & don't of group discussion. | 07 |
| OR | |
| Design a Resume and submit it to the HR (executive) of a company for your placement as engineer in a Company. Write a job application along with the resume | |
| Q3. Answer ALL parts. | |
| (a) If you have to give a presentation which mode of speaking you will choose and why? | 07 |
| OR | |
| Prepare 07slides on the topic 'How to write statement of purpose for higher studies.' | |
| (b) Explain the term Technical Report and its structure. | 07 |
| Q4. Answer ALL parts. | |
| (a) Elaborate PACE model of leadership | 07 |
| (b) Mention the benefits of transformational leadership style. | 07 |
| OR | |
| Differentiate between tact and intelligence. | |
| Q5. Answer ALL parts. | |
| (a) Elaborate different types of personality and their characteristics. | 07 |
| (b) Elaborate Swami Vivekananda's concept of personality development . | 07 |
| OR | |
| What should be the objectives of negotiations? Which personality traits make one a successful negotiator. | |

Subject Code: ICS-401

Roll No.:

2200520100051

B.TECH.

(SEM IV) EVEN SEMESTER EXAMINATION 2023-24

OPERATING SYSTEMS

[TIME: 3 hrs.]

[Max. Marks: 70]

Note: Attempt All Questions. All Question carry equal marks.

- Q1. Answer ALL parts. Marks
- (a) Define the functions of operating systems. Write a note on multi programmed operating systems. 3.5
- (b) Explain the concept of a system call with an example. Mention major categories of System Calls. 3.5
- (c) Explain the layered approach of operating system with diagram 3.5
- OR
- Explain dual mode operation in OS.
- (d) Define context switching in the context of operating systems. Describe the steps involved in a context switch. 3.5

OR

Explain operating system services with respect to user and system.

Q2. Answer ALL parts.

- (a) Calculate average waiting time and average turnaround time for the following snapshot of process using i) SRTF ii) RR scheduling (quantum slice 2 unit) Also draw the Gantt Chart 7

	Arrival Time	Burst Time
P1	0	7
P2	1	3
P3	2	1
P4	3	3

- (b) Define deadlock. How does deadlock avoidance differ from deadlock prevention? 7
Write deadlock avoidance algorithm.

OR

Snapshot of a system is given as

	Allocation	Max	Available
	A B C	A B C	A B C
P0	0 0 2	0 0 4	1 0 2
P1	1 0 0	2 1 0	
P2	1 3 5	1 3 7	
P3	6 3 2	8 4 2	
P4	1 4 3	1 5 7	

Using banker's algorithm answer the following questions:

- (i) What is the content of the matrix need?
- (ii) Whether the system is in safe state or not. If state is safe how it is possible for all processes to complete.
- (iii) If process P2 request for resources (0,0,2), can the request be immediately granted?

5.25
2.75
P1 P2 P3 P4 P2 P4 P4
9
5.5

Q3. Answer ALL parts.

- (a) Free memory holes of sizes 50 KB, 400 KB, 130 KB, 300 KB, 150 KB, and 70 KB (in that order) are available. The processes of size $A = 230$ KB, $B = 180$ KB, $C = 130$ KB, $D = 120$ KB, $E = 200$ KB (in order) is to be allocated. How processes are placed in first fit, best fit, worst fit. Calculate internal as well as external fragmentation.
- (b) Describe the working of the Demand Paging mechanism. How does it help in the efficient utilization of memory?

OR

How many page faults would occur for the following replacement algorithms, assuming three page frames? All the frames are initially empty, page reference string is as: 4, 7, 6, 1, 7, 6, 1, 2, 7, 2, 6, 1, 2

- (i) FIFO page replacement
- (ii) LRU page replacement
- (iii) OPTIMAL page replacement

Q4. Answer ALL parts.

- (a) (i) Define critical sections? Write key requirements that any solution to the critical section problem must satisfy.
- (ii) What is the main objective of process synchronization in a multi-process system?
- (b) Explain how semaphore provide solution of Readers writers problem.

OR

- (i) Explain binary semaphores with the help of suitable example.
- (ii) The following two functions P1 and P2 that share a variable B with an initial value of 2 execute concurrently.

P1() {	P2() {
$C = B - 1;$	$D = 2 * B;$
$B = 2 * C;$	$B = D - 1;$
}	}

Find the number of distinct values that B can possibly take after the execution .

Q5. Answer ALL parts.

- (a) Define file allocation in the context of an operating system. Describe the types of file allocation methods with a suitable diagram.
- (b) What is disk scheduling? Explain the C-SCAN scheduling by giving an example.

OR

The disk queue in the request for input output to block on cylinders 23, 89, 34, 132, 42, 187

Currently the head is at 100. Total 200 cylinders (0-199). Compute the total head movement for the following disk scheduling algorithms: FCFS, SSTF, SCAN and C-SCAN.

Subject Code: ICS 402

Roll No.:

2200520100061

B.TECH.**(SEM IV) EVEN SEMESTER EXAMINATION 2023-24****OBJECT ORIENTED PROGRAMMING USING JAVA****[TIME: 3 hrs.]****[Max. Marks: 70]****Note: Attempt All Questions. All Question carry equal marks.**

- | | | |
|------------|---|--------------|
| Q1. | Answer ALL parts. | Marks |
| (a) | Write and explain the important object-oriented characteristics in Java in brief. | 3.5 |
| (b) | Explain the compilation process of a Java program. | 3.5 |
| (c) | Classify data types in Java along with their size and types of values. | 3.5 |

OR

- | | | |
|-----|---|-----|
| | Write the backslash characters used in Java along with their description. | |
| (d) | Explain the structure of the for loop. Explain its different variants. | 3.5 |

OR

Describe exiting from a loop using the break statement.

- | | | |
|------------|---|---|
| Q2. | Answer ALL parts. | |
| (a) | Describe the important types of inheritance applicable in Java. Explain how it helps to create new classes quickly. | 7 |
| (b) | Define a superclass Parent with a method showMessage(). Create a subclass Child that tries to override showMessage(). Mark showMessage() in Parent as final and demonstrate the compilation error that occurs when trying to override it. | 7 |

OR

Create a class Calculator with two methods with the same name but different parameters add(int a, int b) and add(int a, int b, int c). Write a main method to demonstrate calling both methods. Explain the outputs.

- | | | |
|------------|---|---|
| Q3. | Answer ALL parts. | |
| (a) | Define the stream classes. Describe and differentiate between Byte stream classes and character stream classes. | 7 |
| (b) | Write the process of file creation in Java. Write a program in Java for copying the characters from one file into another file and explain. | 7 |

OR

Write common Java exceptions with descriptions in tabular form. Apply with descriptions try, catch, and finally keywords in a non-trivial example.

- | | | |
|------------|---|---|
| Q4. | Answer ALL parts. | |
| (a) | Write the importance of threads in Java programming. How a thread can be created? Describe the different states of a thread and explain the transition from one state to another. | 7 |
| (b) | Describe the process of inter-thread communication. Write a program in Java in which a thread takes input from another thread. | 7 |

OR

Define Lamda expressions in Java. Describe the following types of Lamda expression with: (i) Zero parameter (ii) Single parameter (iii) Multiple Parameters.

Q5. Answer ALL parts.

- (a) Describe the interfaces in the Java collection framework. 7
- (b) Explain the `AbstractList` class and `LinkedList` class in Java collections with self-explanatory programs. 7

OR

Explain the creation and updation of a linked list by writing a program in Java.

B.TECH.
(SEM IV) EVEN SEMESTER THEORY EXAMINATION, 2023-24
THEORY OF AUTOMATA AND FORMAL LANGUAGES

Time: 3 Hrs

Maximum Marks: 70

Note:

- (i) Answer ALL questions. All questions carry equal marks.
(ii) Notations/ Symbols/ Abbreviations used have usual meaning.

- Q1. Answer ALL parts. Marks
- (a) Define deterministic finite automata (DFA) and the language accepted by DFA. Illustrate the definitions with suitable example. 3.5
Explain whether language accepted by any DFA is always same as the language recognized by that DFA? Explain.
- (b) If $L_1 \subseteq L_2$ and L_2 is not regular then L_1 is not regular. Prove the statement or give counter example. 3.5
- (c) Discuss the Chomsky hierarchy of languages. 3.5

OR

- (i) Write the regular expression for the set of all strings of 0's and 1's containing no more than one occurrence of 11.
- (ii) Simplify the regular expression $r_1(r_1^* r_1 + r_1^*) + r_1^* (r_1 + \epsilon)^*$
- (d) Construct a minimum state DFA equivalent to following automata where state A is the start state and states C, F, I as final states. 3.5

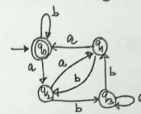
Present State	Next State	
	Input 0	Input 1
→A	B	E
B	C	F
C	D	H
D	E	H
E	F	I
F	G	B
G	H	B
H	I	C
I	A	E

OR

Obtain a minimum state DFA which accepts all those strings over $\Sigma = \{1, 5, 7, 8\}$, which if treated as decimal number is divisible by 3

Q2. Answer ALL parts.

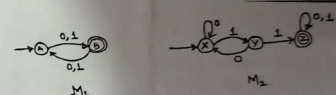
- (a) Obtain the regular expression for the following finite automata. 7



- (b) State the pumping lemma for regular languages and hence use it to prove that the given Language L is not regular. 7
 $L = \{a^m \mid m \text{ is not a perfect cube}\}$

OR

If M_1 and M_2 be the finite automata recognizing the languages L_1 and L_2 respectively. Draw the finite automaton recognizing the language $L_1 \cup L_2$.



Q3. Answer ALL parts.

- (a) (i) What do you understand by ambiguous grammar? Find out whether a CFG with following productions is ambiguous or not. 7
 $S \rightarrow S(S) \mid \epsilon$
(ii) Write a context free grammar which generates all the strings made up of 0's and 1's having more 0's than 1's.
(iii) Show that complement of a context free language is not necessarily a context free language.

- (b) (i) What do you understand by left recursion in a CFG? Illustrate through a suitable example how left recursion can be removed from any CFG having left recursion. 7
(ii) Simplify the given CFG to obtain an equivalent CFG with no useless symbols.

$S \rightarrow ABC \mid BaB$
 $A \rightarrow aA \mid BaC \mid aaa$
 $B \rightarrow bBb \mid a$
 $C \rightarrow CA \mid AC$

OR

Convert the following grammar into Greibach Normal Form (GNF).

$S \rightarrow S+T \mid T$
 $T \rightarrow T * F \mid F$
 $T \rightarrow (S) \mid a$

Q4. Answer ALL parts.

- (a) (i) Give a formal definition of deterministic PDA. What do you understand by Instantaneous Description (ID) of Pushdown automata? 7
- (ii) Construct a Push Down Automata (PDA) which accepts the language L defined as follows:
 $L = \{ 0^m 1^n 2^p \mid p = m + n + 2; m, n, p \text{ are natural numbers} \}$
- (b) Obtain the equivalent CFG that generates the same language L which is accepted by the given PDA $M = (\{q_0, q_1\}, \{0, 1\}, \{X, Z_0\}, \delta, q_0, Z_0, \Phi)$ by empty stack. 7

$\delta(q_0, 0, Z_0) = (q_0, XZ_0)$
 $\delta(q_0, 0, X) = (q_0, XX)$
 $\delta(q_0, 1, X) = (q_1, X)$
 $\delta(q_1, 0, X) = (q_1, \epsilon)$
 $\delta(q_1, 1, Z_0) = (q_0, \epsilon)$

OR

Use the CYK algorithm to decide whether the string $aaabba$ is generated by the following grammar or not?

$S \rightarrow AB$
 $A \rightarrow AA \mid AB \mid a$
 $B \rightarrow CC$
 $B \rightarrow b$

OR

Prove that any language accepted by some PDA by final state is also acted by some PDA by empty stack and vice-versa.

Q5. Answer ALL parts.

- (a) Design a Deterministic Turing machine and draw the graphical representation of the Machine for anyone of the following. 7
- (i) Turing Machine that computes the function f given by $f(n) = n/2$; where n is even natural numbers.
- (ii) Turing Machine that accepts all the strings over $\Sigma = \{0, 1\}$ having equal number of 0's and 1's.
- (b) (i) Prove that if both a language L and its complement are recursively enumerable (r.e.) then language L is recursive. 7
- (ii) What do you understand P Class and NP Class of problems. Discuss the significance of the question whether $P = NP$ or not?

OR

- (i) Obtain the solution, if it exists, to the following instances of PCP.

$A = (01, 001, 10), B = (011, 01, 00)$

- (ii) Write short note on Universal Turing Machine.

B. Tech.
(SEM IV) EVEN SEMESTER EXAMINATION 2023-24
PYTHON PROGRAMMING

[TIME: 3 hrs.]

Note: Attempt All Questions. All Question carry equal marks.

[Max. Marks: 70]

- Q1. Answer ALL parts.**
- (a) Write a short note on history and features of python. Marks
3.5
- (b) Define List and Tuple in Python. Differentiate between List and Tuple using example. 3.5
- OR**
- Write the pseudo code for the concatenation, repetition and making sublists from given two lists.
- (c) Write a python code to demonstrate type conversions using int (), float () and str (). 3.5
- OR**
- (d) Explain the precedence and associativity of various operators used in python. 3.5
- What are the various applications of Python? Give examples.
- Q2. Answer ALL parts.**
- (a) Write a program to print the following pattern. 7
- ```

A
ABA
ABCAB
ABCDABC
ABCDEABCD

```
- (b) Mention the different types of loop and control statements allowed in Python and explain each type with suitable examples and flowchart. 7
- OR**
- Create a Python script to multiply two matrices through nested loops and transpose the resulting matrix.
- Q3. Answer ALL parts.**
- (a) (i) Explain various inbuilt methods of String with suitable example of each. 7
- (ii) What will be the output of the given code?
- ```
list("python")
```
- (b) Illustrate the ways of creating the tuple and the tuple assignment with suitable programs. Describe the built-in functions with tuples. 7
- OR**
- (i) Write a python program to check whether a string is a palindrome or not.
- (ii) Write a python program to accept the strings which contain all vowels.
- Q4. Answer ALL parts.**
- (a) Write the syntax to create, open and close a file? Write a Python program to read a file line-by-line. 7
- (b) Write Python Program to Count the Occurrences of Each Word and Also Count the Number of Words in a text file given as input. 7
- OR**
- Write a Python program that reads a text file and changes the file by capitalizing each character of file.
- Q5. Answer ALL parts.**
- (a) Discuss about GUI Programming in Python? Explain how to create GUI application using Tkinter? 7
- OR**
- Explain the following Widgets in Tkinter
- (i) Button (ii) Check button (iii) Frame (iv) Canvas
- (b) (i) Explain how a module is created. Discuss the advantages of using modules. 7
- (ii) Write a program to multiply two matrices using NumPy module.

B.TECH.

(SEM IV) EVEN SEMESTER EXAMINATION 2023-24
PRACTICES AND ETHICAL ISSUES IN RESEARCH

[TIME: 3 hrs.]

[Max. Marks: 70]

Note: Attempt All Questions. All Question carry equal marks. Make suitable assumptions wherever necessary.

Q1 Answer ALL parts.

- (a) Describe the main objectives of conducting research and explain how these objectives align with ethical principles in scientific inquiry.

Marks

7

OR

Define research misconduct. What impact do unethical practices have on the scientific community? Explain.

- (b) What are some common issues in research and how can institutions work to prevent such behavior?

7

Q2 Answer ALL parts.

- (a) Identify and discuss the primary challenges researchers face in avoiding plagiarism. What tools and strategies can be employed to ensure proper citation and originality?
- (b) Describe the process of scientific reading and how it differs from general reading. What techniques can researchers use to critically evaluate and synthesize information from scientific literature?

7

7

OR

Describe how design thinking methods can be employed to avoid plagiarism in academic writing. What specific strategies can researchers use to ensure originality in their work?

Q3 Answer ALL parts.

- (a) What are the key principles and criteria that define good research practice, and how do they contribute to maintaining research integrity?

7

OR

Discuss the importance of proper data management in research. What best practices should researchers follow to ensure data integrity and confidentiality throughout the research lifecycle?

- (b) Differentiate between secrecy and confidentiality in the context of handling research data with suitable examples.

7

Q4 Answer ALL parts.

- (a) Discuss the process of formulating a research problem. What are the key considerations researchers must take into account to ensure their problem statement is clear, relevant, and feasible?
- (b) Explain the responsibilities of a researcher with respect to good research.

7

7

OR

Describe the steps involved in preparing and presenting research results in project proposals.

Q5 Answer ALL parts.

- (a) Researchers and organizations work together to promote creativity and innovation. Elaborate.

7

OR

What role do style manuals (e.g., APA, MLA, Chicago, IEEE) play in maintaining consistency and credibility in academic writing?

- (b) Differentiate falsification, manipulation and plagiarism with suitable example.

7